Tutorial 9

Options Chain Basics: Strike, Expiration, Calls, Puts, Volume, and Open Interest

A student-ready module for reading option-chain information before trading options.

Module purpose

Prepared from the April 21, 2026 class transcript. Original discussion anchor: the GLD/SPY-style option-chain walkthrough on asset price versus strike price, DTE, calls, puts, volume, open interest, and why students should analyze options chains before trading options. Educational material only - not investment advice.

Ideal length	Audience	Student output
90 minutes	Options-analysis beginners	Annotated option chain + observation log

1. Lesson snapshot

What the live conversation was really about

The class used GLD and a SPY-like options chain to teach the first layer of options-chain analysis: asset price versus strike price, expiration or DTE, calls versus puts, volume, open interest, and basic liquidity. The purpose is not to teach students to trade options immediately. The purpose is to teach them to read the chain as market information.

Conversation idea	Student-ready correction	Why it matters
Strike price and asset price were compared.	The asset price is the current price of the underlying. The strike is the exercise price written into the option contract.	Students must not confuse the price of GLD/SPY with the price of the option contract.
DTE means days to expiration.	0DTE expires today, 1DTE expires next trading day, and longer DTE gives more time value.	Expiration changes the speed of risk, especially near the money.
Volume and open interest were used to judge participation.	Volume is contracts traded during the period. Open interest is the number of open contracts still active after opening/closing activity is processed.	Open interest is not simply yesterday's volume; it changes depending on whether trades open or close positions.
Put/call skew or ratio was used to filter noise.	Compare calls and puts within the same underlying and expiration first. Then ask whether the difference is meaningful or just liquidity/noise.	A ratio is context, not a stand-alone buy or sell signal.

2. Learning objectives

- Identify the parts of a basic option chain: underlying, expiration, strike, call side, put side, bid, ask, last, volume, and open interest.
- Explain the difference between asset price, strike price, and option premium.
- Use DTE to organize the chain before comparing contracts.
- Use volume and open interest as liquidity/participation clues without pretending they reveal direction by themselves.
- Create a paper-only option-chain observation log.

3. Contract basics in plain English

Term	Beginner definition	Common mistake
Underlying	The asset the option is based on, such as GLD, SPY, or a stock.	Thinking the option is the same thing as owning the underlying.
Call	A contract giving the buyer the right, not obligation, to buy the underlying at the strike price under the contract terms.	Assuming all call volume is automatically bullish.
Put	A contract giving the buyer the right, not obligation, to sell the underlying at the strike price under the contract terms.	Assuming all put volume is automatically bearish.
Strike	The exercise price written into the option contract.	Calling the strike the same thing as the current market price.
Expiration	The date after which the option no longer has rights. DTE counts days to that expiration.	Ignoring how quickly risk changes near expiration.
Premium	The option's market price, usually quoted per share for standard equity options.	Forgetting that one standard equity option usually controls 100 shares.

4. How to read the chain: six passes

Pass	Question	Action
1. Underlying	What asset is this chain attached to?	Confirm ticker, exchange, and live underlying price.
2. Expiration	Which DTE are we studying?	Choose one expiration first. Do not mix 0DTE, 1DTE, and weekly contracts in the same comparison.
3. At-the-money area	Which strike is closest to the underlying price?	Mark ATM, then one or two strikes above and below.

Pass	Question	Action
4. Call/put comparison	Where is volume heavier?	Compare call and put volume at the same expiration and near the same strike region.
5. Liquidity	Can a student enter/exit without huge slippage?	Check bid-ask spread, volume, and open interest.
6. Context	Does the chain confirm a chart level or catalyst?	Connect chain data to support/resistance, VWAP, news, and scheduled events.

Teaching phrase

First choose the expiration. Then find the strike neighborhood. Then compare call/put activity. Chain analysis fails when students mix different expirations and pretend they are one data set.

5. A simplified option-chain layout

Real platforms differ, but most chains organize calls on one side, puts on the other, and strikes in the middle. The exact columns can change, so students should learn the concepts rather than memorize one app layout.

Calls: OI	Calls: Vol	Bid	Ask	Strike	Bid	Ask	Puts: Vol	Puts: OI
1,850	3,200	4.10	4.25	270	2.05	2.18	1,400	2,600
2,400	5,800	2.75	2.90	275	3.25	3.40	2,250	3,100
4,900	9,100	1.55	1.65	280	5.20	5.45	3,900	5,600

Example interpretation: if the underlying is near 278, the 280 strike is close to at-the-money. The student should compare call and put volume near 275-280, then check whether bid-ask spreads are reasonable. This table is an illustration, not live data.

6. Volume vs. open interest

Metric	What it means	How students use it	What it does NOT prove
Volume	How many option contracts traded during the selected period, often the current trading day.	Shows current activity and whether a strike is active today.	It does not prove whether buyers or sellers are smarter, nor does it prove future direction.
Open interest (OI)	The number of open option contracts still active after opening and closing activity is processed.	Shows existing participation and can help locate liquidity or popular strikes.	It is not the same as yesterday's volume and does not reveal bullish/bearish direction by itself.
Bid-ask spread	The difference between the highest bid and lowest ask currently quoted.	Shows execution cost and liquidity quality.	A tight spread does not guarantee the trade is good; it only means entry/exit may be cleaner.

Open interest correction

Open interest can rise, fall, or stay unchanged depending on whether trades are opening or closing positions. For every option contract there is both a buyer and a seller, so OI alone does not tell you which side is "right."

7. Put/call ratio and skew - beginner version

A simple put/call volume ratio is calculated as put volume divided by call volume. A ratio above 1 means more put volume than call volume in the slice being measured; below 1 means more call volume. But the ratio must be defined. Is it for one strike? one expiration? the whole chain? index options? ETF options?

Comparison	Formula / observation	Beginner interpretation
Same strike, same expiration	Put volume at strike / call volume at strike	Useful for local interest, but can be noisy.
Whole expiration	Total put volume for expiration / total call volume for expiration	Better for expiration-level mood, but still not a signal alone.
Open interest ratio	Total put OI / total call OI	Shows inventory of open contracts, not current-day flow.

Comparison	Formula / observation	Beginner interpretation
Skew	Relative richness or activity across puts vs calls	Advanced. Students should first master volume, OI, strike, and expiration.

8. Beginner safety rules

- Use option chains for analysis before trading options. Chain literacy comes before option execution.
- Paper observe 0DTE contracts; do not trade them until risk, spread, assignment, decay, and position sizing are mastered.
- Do not use market orders in wide-spread options. The spread is a real cost.
- Do not buy the strike with the biggest volume automatically. Ask why the volume exists and whether liquidity is clean.
- Record the underlying price, expiration, ATM strike, highest call volume, highest put volume, highest OI, and spread before forming an opinion.

9. Paper-only observation log

Field	Student entry
Date/time	
Underlying and price	
Expiration / DTE	
ATM strike	
Highest call volume strike	
Highest put volume strike	
Highest open interest strike	
Bid-ask spread near ATM	
Chart level nearby	
What changed after 30 minutes?	

10. Classroom exercise

Give students a screenshot of a real option chain. They should not place trades. They should annotate the chain and explain what each column tells them.

Task	Correct answer should include
Mark the underlying price.	Asset price is separate from option premium.
Circle the ATM strike.	The strike closest to the current underlying price.
Pick one expiration.	Students do not mix expirations.
Find highest call and put volume.	Volume by strike and expiration, not a vague total.
Find highest open interest.	OI by strike and expiration; explain it is open contracts, not yesterday's volume.
Judge liquidity.	Bid-ask spread, volume, and OI together.

11. Mini-assessment

- Define asset price, strike price, and option premium in one sentence each.
- What does 1DTE mean? How is it different from 0DTE?
- Why is open interest not the same as yesterday's volume?
- Why can high call volume fail to push the underlying higher?
- What three data points would you check before deciding an option is liquid?

Source-backed reading list

Use these sources to complement the transcript. Students should prefer primary filings, regulator pages, and official product documentation before social media or summaries.

[1] OIC, Options Basics FAQ - definition of stock options, calls, puts, strike price, and expiration concepts.

[2] OIC, "Open Interest: Why It Matters" - explains active open contracts and how open interest can increase, decrease, or remain unchanged based on opening/closing activity.

[3] OCC, "Characteristics and Risks of Standardized Options" - official disclosure document for exchange-traded options and risk characteristics.

[4] SEC Investor.gov, "An Introduction to Options" - beginner explanation of calls, puts, strike price, and option risks.

[5] Cboe Options Institute, Options Basics and Glossary - listed options education, liquidity, expiration, and strike terminology.

[6] OIC Market Data resources - source for volume, open interest, series, trading data, and contract adjustment information.

Instructor reminder

The objective of this module is chain literacy, not options trading. Students should be able to explain the chain before they are allowed to discuss real-money option strategies.